

Mapping for a Cylindrical Micro-Resistive Well Detector

1. Group Members:

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2. Faculty Advisors:

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3. Clients:

Pietro Iapozzuto- Physics department

Dr. Marcus Hohlmann- Physics department

4. Milestone 4 Task Progress Matrix:

Task	Completion	Alejandro	Summer	TO DO
I. Find correct cylindrical mapping	90%	90%	0%	Current reconstruction shows clear beam but can't confirm until compare with trackers
II. Create efficiency plots	80%	0%	80%	Need to apply clustering and iterative process for shifting the offset
III. Compare cylindrical detectors and trackers	20%	10%	10%	Need to compare detector data to tracker data using the calculated trajectories

5. Discussion of task progress for M4

I. Find correct cylindrical mapping

An optimized and mathematically sound cylindrical reconstruction program was used to test various mapping functions. The AMORE-SRS software that originally converts raw files to root files was also investigated to gain more insight on how the data is being preprocessed. Individual strip hit occupancies were also graphed and evaluated to examine patterns as well as anomalies in the data. The resulting reconstruction was then compared visually to the tracker reconstructions.

II. Create efficiency plots

An exclusive residual analysis pipeline was generated which consists of calculating the trajectories of hits, predicting x-y coordinates of hits based on their z position, comparing the predictions to their measured position on the excluded tracker/detector, and displaying visuals/statistics on their residuals. The average residual value per detector

was added back to each hit position to account for any offset. However, this process of shifting the data will need to be applied iteratively to ensure proper convergence. Additionally, clustering should be applied beforehand so positions aren't determined via raw strip values.

III. Compare cylindrical detectors and trackers

The efficiency pipeline was tested on the trackers alone to be used as a baseline. Although, there is more cleaning to be done on the detector 5 data before we test its efficiency. On the other hand, the detector 5 beam reconstruction was compared visually to the other four tracker reconstructions and there was a strong correlation. Furthermore, comparing residuals from different mapping configurations on the detector 5 will provide more concrete confirmation of success.

6. Discussion of team member contribution

Alejandro- Tried different mappings for detector 5, eventually finding a candidate mapping that produced the expected line. Did reconstruction of the trackers and cylindrical mapping. Explored AMORE-SRS software to gain more insight on how the data is being preprocessed. Graphed individual strip hit occupancies to data anomalies.

Summer- Created pipeline to generate the trajectories of hits. Developed a program that predicts the position of a hit. Displayed metrics for analyzing the metrics of such residuals. Performed tracker position verification using the exclusive residual analysis pipeline. Applied the calculated average residual shift to account for any offset

7. Milestone 5 Task Matrix

Task	Alejandro	Summer
I. Apply clustering and iterative offset shift before reconstructing	50%	50%
II. Automate beam reconstruction pipeline for multiple root files	50%	50%
III. Do efficiency plots and other requested graphs for analysis	50%	50%

8. Discussion of tasks planned for M5

I. Apply clustering and iterative offset shift before reconstructing

Feedback from our clients revealed strip clustering is a very crucial preprocessing step we need to apply instead of comparing raw data from the strips. Furthermore, the clustering neighborhood size will need to be dynamic for each event for the reconstruction and validation. Additionally, the iterative offset shifting algorithm will also need to be performed on the trackers before a true trajectory is calculated.

II. Automate beam reconstruction pipeline for multiple root files

Use multiple root files to do beam reconstruction on trackers and the cylindrical detector. To be able to verify the mapping in multiple datasets, and be able to gather statistics on the experiments.

III. Do efficiency plots and other requested graphs for analysis

Create efficiency plots to verify the accuracy of the cylindrical detector and its mapping, automating the generation of other performance statistics. Such graphs could possibly include: histograms of the total cluster charge, double Gaussian fit on the residuals, strip multiplicity distributions, or other quality metrics. More graphs and analysis will be provided at the request of the physicists and/or as the need arises.

9-10. See faculty meetings and feedback below

11. Dates and Meetings with Faculty:

1. Sept 8th, 2026
2. Sept 16th, 2026
3. Sept 18th, 2026
4. Sept 21st, 2026
5. Sept 25th, 2026
6. Sept 28th, 2026

12. Faculty feedback on tasks for M4

- Use clustering for events with multiple strip activations for the reconstruction and verification
- Make the clustering neighborhood size dynamic for each event for the reconstruction and validation
- Find a dynamic adc threshold for each APV, because multiple APVs have different levels of noise
- Apply an iterative approach to shifting the tracker data and accounting for offset
- Fit a double Gaussian on the residuals for better analysis

13. Approval from Faculty Advisor

- "I have discussed with the team and approve this project plan. I will evaluate the progress and assign a grade for each of the three milestones."
- Signature: _____ Date: _____